

Problem set 3

Exponential maps and robot kinematics

Modern Robotics

2013

Before you start, please note: while the numbers in these exercises were chosen such that you can complete the exercises by hand, it might be less work if you use the set of MATLAB scripts that was made available. Check the file `examples.m` for how to work with those functions—they could come in really handy in especially questions 3 and 4.¹

1. Matrix exponentials

In Brockett’s exponential formula (see question 3) we make use of the tilde form transformation of twists and the property of the matrix exponential given in (1). Verify that we may use this by proving the property.

$$e^{HAH^{-1}} = He^AH^{-1} \quad (1)$$

Hint: start with the definition of the matrix exponential.

2. Rodriguez’ formula

Rotation matrices can be quite difficult to calculate, especially when trying to remember which sine—or sign—goes where. However, in last homework set we saw that the time derivative of a rotation matrix² can be easily calculated.

The resulting differential equation has a solution as in (2), where the exponential $e^{\tilde{\omega}}$ can be readily calculated with Rodriguez’ formula as given in the slides.

$$R_i^j(t) = e^{\tilde{\omega}_i^{j,j}t} \quad (2)$$

Suppose that we execute a rotation by applying a rotational velocity of $\omega_i^{j,j} = (\frac{\pi}{6}, 0, 0)^\top \text{ rad s}^{-1}$ for 2 s; what would be the final rotation matrix R_i^j ?

Hint: To use Rodriguez’ formula, you’ll need to express $\tilde{\omega}$ as a unit rotation $\hat{\omega}$ with $\|\omega\| = 1$ —choose θ such that $\hat{\omega}\theta = \tilde{\omega}t$.

¹Although Rodriguez’ formula is also in the functions, try to do question 2 “by hand”—you can use MATLAB, but don’t use the supplied function.

²Alright, the question was on a homogeneous matrix, but the result holds if you substitute H with R and $\tilde{\omega}$ for $\tilde{T}$.

3. Brockett's exponential formula

Consider the crane depicted in Figure 1, which is a type often seen in for example harbours: it consists of a **base** that can ride on a **track** (q_1 [m]) and a **pillar** of fixed height with a rotating **boom head** (q_2 [rad]). The **cable** hangs from a pulley that can move along the **boom** (q_3 [m]) and of course the hook can be lowered (q_4 [m]).

Use Brockett's exponential formula to express the orientation (position+rotation) of Ψ_4 , attached to the hook, with respect to Ψ_0 , the inertial frame attached to the start of the track, when $q = (q_1, q_2, q_3, q_4)^\top = (3 \text{ m}, -\frac{\pi}{2} \text{ rad}, 2 \text{ m}, 3 \text{ m})^\top$.

Hint: First find the unit twists $\tilde{T}_i^{0,i-1}$ and then calculate each matrix exponential as explained in the slides.

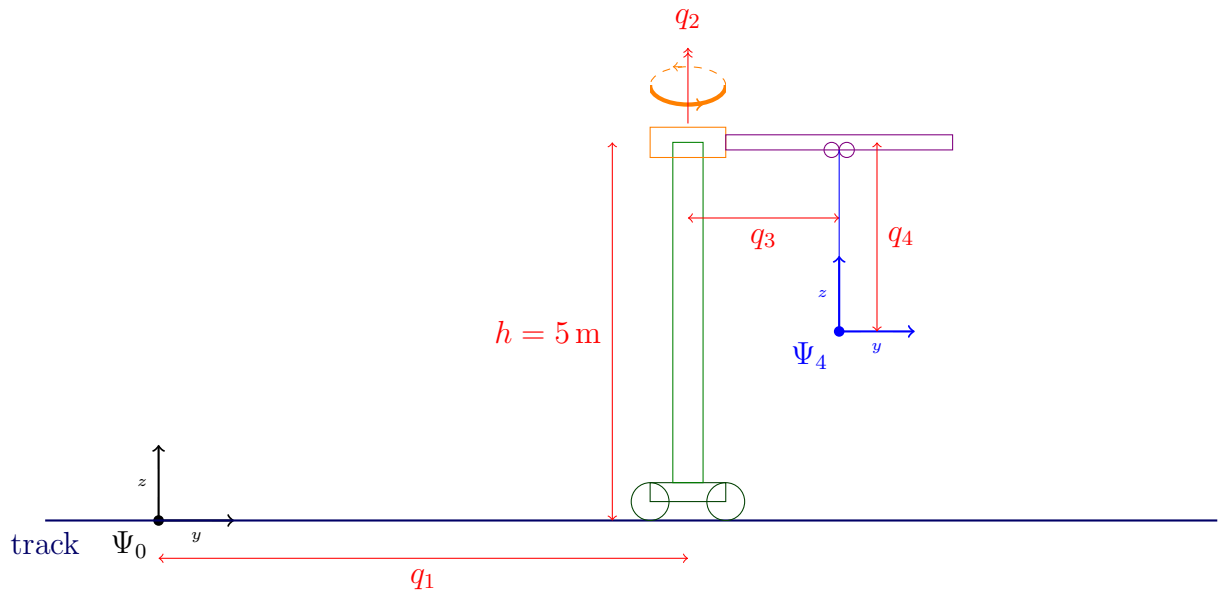


Figure 1: The crane considered in questions 3 and 4. Note that the joint positions q_i are drawn arbitrarily—with the exception of q_2 , which is 0 rad as drawn and hence both frames' $\hat{x}$ axes point out of the paper right now.

4. Geometric Jacobian

As a crane operator it is very important to know the speed of the load (hook), because if it moves too fast, it might bump into something and cause large damage. Of course the operator only has speed control over the 4 joints, so he needs your help to calculate the end-effector twist $T_4^{0,0}$.

Given the configuration in question 3, what is the instantaneous end-effector twist $T_4^{0,0}$ when the joint velocities are $\dot{q} = (2 \text{ m s}^{-1}, -1 \text{ rad s}^{-1}, 0 \text{ m s}^{-1}, 1 \text{ m s}^{-1})^\top$?

Hint: First construct the Geometric Jacobian $J(q)$, of which T_2 and T_3 depend on the configuration q .

Note that the in result $T_4^{0,0} = (\omega, v)^\top$ the velocity v is defined in a special, geometric way rather than using $\dot{r}$ (see page 33 of Les2.pdf), so don't be confused by numbers in the outcome that may seem strange—try transforming the twist to Ψ_4 for a more intuitive representation.³

³Using $\text{Ad}_{H_0^4}$ of course.